COMSATS UNIVERSITY ISLAMABAD
SAHIWAL CAMPUS

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Sign Bridge
(AI-Sign Language Translator)

SUPERVISED BY
Dr Yawar Abbas Abid

ZARYAB NADEEM (FA22-BCS-124)
MUHAMMAD UMAIR RANA (FA22-BCS-198)
MUHAMMAD UZAIR (FA22-BCS-225)

WORKING

The user performs a sign language gesture in frontof the mobile camera, where the application captures the live video feed. Using MediaPipe Hand Landmark Detection, the system identifies the hand and extracts 21 key hand landmarks, which are converted into feature vectors. These features are then processed by a trained TensorFlow Lite AI model to recognize the corresponding sign accurately. The predicted sign is instantly translated into readable text. It can also be converted into speech, enabling fast, accurate, and real-time communication between deaf or mute individuals and the hearing community.

1 PROBLEM STATEMENT

Communication between deaf or mute individuals and people unfamiliar with sign language is difficult. Existing solutions are expensive, require internet access, or depend on human interpreters.

2 PROPOSED SOLUTION

SignBridge is an AI-powered mobile app that recognizes sign language and instantly converts it into text and speech for seamless communication.

3 METHODOLOGY

The system captures hand gestures, extracts 21 hand landmarks using MediaPipe, and uses a TensorFlow Lite model to translate signs into text and speech in real time.

4 TOOLS & TECHNOLOGIES

- **Frontend:** Flutter, Dart
- **AI & Machine Learning:** TensorFlow Lite, MediaPipe Hands, OpenCV
- **Programming:** Python
- **Development Tools:** Android Studio, Visual **Studio Code**